

For this assignment, I implemented three Langton's Ants types, each resulting in different visual behaviors. The first of these ant behaviors was the original ant behavior provided by the main seagulls example. For the second ant behavior, I wanted to go for a more random visual effect. I created a variable that generates a random value between 0.0 and 1.0 (<https://discussions.unity.com/t/generate-random-float-between-0-and-1-in-shader/728060>), and then influenced the ant direction and pheromone drawing based on whether this random value is greater than 0.5 or the opposite. The ant direction change and pheromone drawing has a similar format to the first ant behavior, except for what value is used to decide the altered ant state (pheromone for the first behavior vs. random value for the second behavior). Visually, the second behavior has a more varied result, and the ants typically will go further away from their starting point. For the third behavior, I wanted to experiment a bit more with trig functions, and I also wanted to try manipulating the ant position instead of the ant direction. Instead of increasing the ant direction by 0.25 if the pheromone value is 0, this third behavior increases the ant position by the cosine of the current ant position. When the pheromone value is 1, it decreases the ant position by the cosine of the current ant position. This behavior results in an interesting dotted line effect, where the ants will continually go down the screen (appearing back at the top of the screen after going too far off the bottom) and leave a trail of dotted or dashed lines behind them. These lines continually change as the ants overlap their lines again and again.